

Inference and Testing for Randomised Experiments

Ryan T. Moore

American University

2026-07-22

Table of contents I

Preliminaries

Randomisation (Design-based) Inference

Preliminaries

Preliminaries

- ▶ Accept GitHub invitation (click link in email)
- ▶ Upload `.txt` or `.R` or `.qmd` or `.Rmd` of solutions
(`.pdf`, but comments)
- ▶ How `.qmd`/`.Rmd` works (more to come?)

Preliminaries

- ▶ Accept GitHub invitation (click link in email)
- ▶ Upload `.txt` or `.R` or `.qmd` or `.Rmd` of solutions
(`.pdf`, but comments)
- ▶ How `.qmd`/`.Rmd` works (more to come?)
- ▶ PS1 for today

Preliminaries

- ▶ Accept GitHub invitation (click link in email)
- ▶ Upload `.txt` or `.R` or `.qmd` or `.Rmd` of solutions
(`.pdf`, but comments)
- ▶ How `.qmd`/`.Rmd` works (more to come?)
- ▶ PS1 for today
- ▶ PS2 for Thursday

Preliminaries

- ▶ Accept GitHub invitation (click link in email)
- ▶ Upload `.txt` or `.R` or `.qmd` or `.Rmd` of solutions
(`.pdf`, but comments)
- ▶ How `.qmd`/`.Rmd` works (more to come?)
- ▶ PS1 for today
- ▶ PS2 for Thursday
- ▶ Work together, but every keystroke your own

Preliminaries

- ▶ Accept GitHub invitation (click link in email)
- ▶ Upload `.txt` or `.R` or `.qmd` or `.Rmd` of solutions
(`.pdf`, but comments)
- ▶ How `.qmd`/`.Rmd` works (more to come?)
- ▶ PS1 for today
- ▶ PS2 for Thursday
- ▶ Work together, but every keystroke your own
- ▶ Ask questions in the Slack

Randomisation (Design-based) Inference

A volunteer?

A volunteer?

The task: select the 2 folders with messages

A volunteer?

The task: select the 2 folders with messages

- ▶ What is our baseline expectation/model for this process?

A volunteer?

The task: select the 2 folders with messages

- ▶ What is our baseline expectation/model for this process?
 - ▶ “No x-ray vision. No ESP. Effect of messages on choice = 0.”

A volunteer?

The task: select the 2 folders with messages

- ▶ What is our baseline expectation/model for this process?
 - ▶ “No x-ray vision. No ESP. Effect of messages on choice = 0.”
- ▶ What is an alternative?

A volunteer?

The task: select the 2 folders with messages

- ▶ What is our baseline expectation/model for this process?
 - ▶ “No x-ray vision. No ESP. Effect of messages on choice = 0.”
- ▶ What is an alternative?
 - ▶ “Some way to detect messages. Message location $\rightarrow$ choice.”

A volunteer?

The task: select the 2 folders with messages

- ▶ What is our baseline expectation/model for this process?
 - ▶ “No x-ray vision. No ESP. Effect of messages on choice = 0.”
- ▶ What is an alternative?
 - ▶ “Some way to detect messages. Message location $\rightarrow$ choice.”

A volunteer?

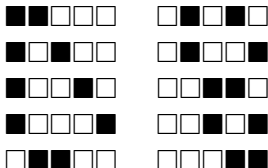
The task: select the 2 folders with messages

- ▶ What is our baseline expectation/model for this process?
 - ▶ “No x-ray vision. No ESP. Effect of messages on choice = 0.”
- ▶ What is an alternative?
 - ▶ “Some way to detect messages. Message location $\rightarrow$ choice.”

Select!

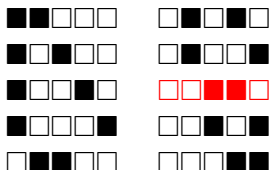
Randomisation Inference

The possible choices:



Randomisation Inference

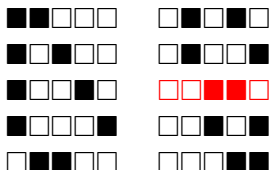
The possible choices:



► You chose ____ and _____. Let X = number found.

Randomisation Inference

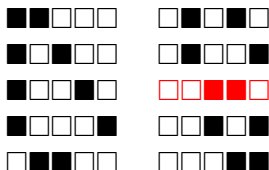
The possible choices:



- ▶ You chose _____ and _____. Let X = number found.
- ▶ What was $P(X \geq 2 | \text{no ESP})$?

Randomisation Inference

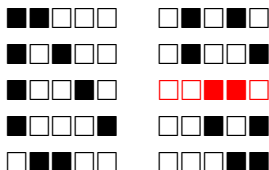
The possible choices:



- ▶ You chose ____ and _____. Let X = number found.
- ▶ What was $P(X \geq 2 | \text{no ESP})$? $\frac{1}{10} = 0.1$

Randomisation Inference

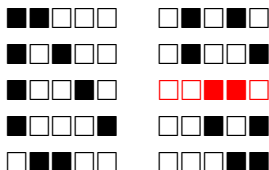
The possible choices:



- ▶ You chose ____ and _____. Let X = number found.
- ▶ What was $P(X \geq 2 | \text{no ESP})$? $\frac{1}{10} = 0.1$
- ▶ What was $P(X \geq 1 | \text{no ESP})$?

Randomisation Inference

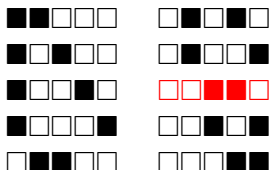
The possible choices:



- ▶ You chose _____ and _____. Let X = number found.
- ▶ What was $P(X \geq 2 | \text{no ESP})$? $\frac{1}{10} = 0.1$
- ▶ What was $P(X \geq 1 | \text{no ESP})$? $\frac{7}{10} = 0.7$

Randomisation Inference

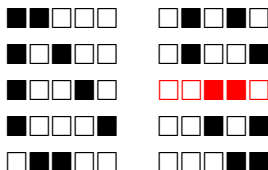
The possible choices:



- ▶ You chose ____ and _____. Let X = number found.
- ▶ What was $P(X \geq 2 | \text{no ESP})$? $\frac{1}{10} = 0.1$
- ▶ What was $P(X \geq 1 | \text{no ESP})$? $\frac{7}{10} = 0.7$
- ▶ What is “prob result at least this extreme, given model of no effect”?

Randomisation Inference

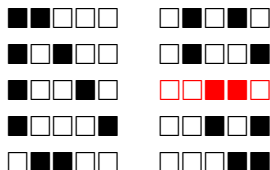
The possible choices:



- ▶ You chose ____ and _____. Let X = number found.
- ▶ What was $P(X \geq 2 | \text{no ESP})$? $\frac{1}{10} = 0.1$
- ▶ What was $P(X \geq 1 | \text{no ESP})$? $\frac{7}{10} = 0.7$
- ▶ What is “prob result at least this extreme, given model of no effect”?
- ▶ Definition of p -value!

Randomisation Inference

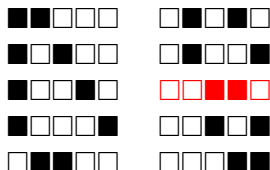
The possible choices:



- ▶ You chose ____ and _____. Let X = number found.
- ▶ What was $P(X \geq 2 | \text{no ESP})$? $\frac{1}{10} = 0.1$
- ▶ What was $P(X \geq 1 | \text{no ESP})$? $\frac{7}{10} = 0.7$
- ▶ What is “prob result at least this extreme, given model of no effect”?
- ▶ Definition of p -value!
- ▶ Valid, exact, with no distributional assumption, no large n

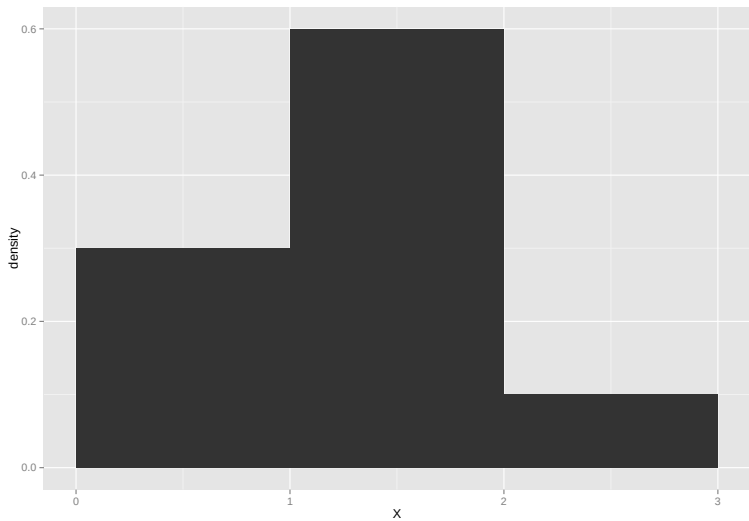
Randomisation Inference

The possible choices:

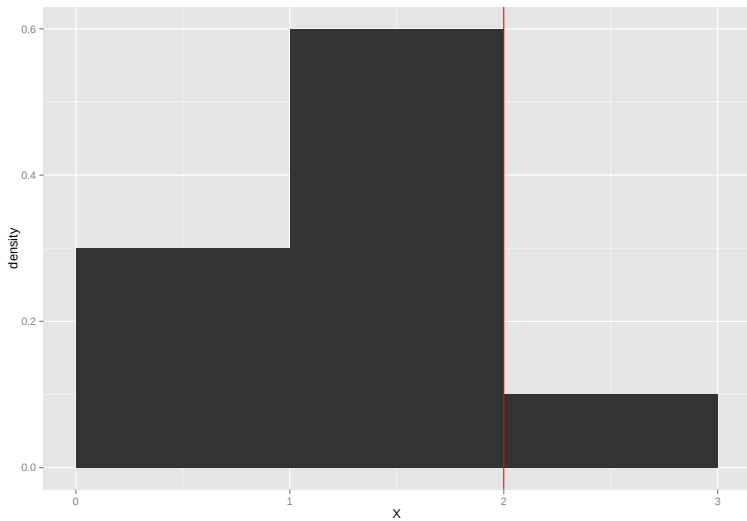


- ▶ You chose ____ and _____. Let X = number found.
- ▶ What was $P(X \geq 2 | \text{no ESP})$? $\frac{1}{10} = 0.1$
- ▶ What was $P(X \geq 1 | \text{no ESP})$? $\frac{7}{10} = 0.7$
- ▶ What is “prob result at least this extreme, given model of no effect”?
- ▶ Definition of p -value!
- ▶ Valid, exact, with no distributional assumption, no large n
- ▶ *Randomisation* creates dist’n of possible numbers correct

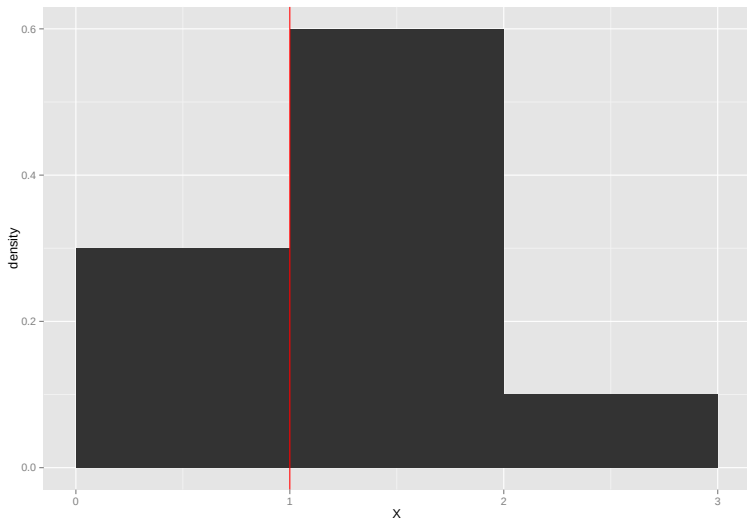
The Randomisation Distribution of X



The Randomisation Distribution of X



The Randomisation Distribution of X



Parametric Null Hypothesis Significance Testing

- ▶ Specify and assume H_0
- ▶ Define H_A
- ▶ Examine reference dist'n (t , χ^2 , ...) under H_0
- ▶ Calculate p -value
- ▶ Compare to some α ; reject H_0 if $p < \alpha$

Randomisation Inference

- ▶ Specify and assume H_0
(sharp null of no treatment effect)

Randomisation Inference

- ▶ Specify and assume H_0
(sharp null of no treatment effect)
- ▶ Define H_A

Randomisation Inference

- ▶ Specify and assume H_0
(sharp null of no treatment effect)
- ▶ Define H_A
- ▶ Create reference dist'n from all possible values of X under H_0
(or at least a big sample of them)

Randomisation Inference

- ▶ Specify and assume H_0
(sharp null of no treatment effect)
- ▶ Define H_A
- ▶ Create reference dist'n from all possible values of X under H_0
(or at least a big sample of them)
- ▶ What prop. of possible values are “at least as extreme as” observed?
 $\leadsto$ p -value!

Randomisation Inference

- ▶ Specify and assume H_0
(sharp null of no treatment effect)
- ▶ Define H_A
- ▶ Create reference dist'n from all possible values of X under H_0
(or at least a big sample of them)
- ▶ What prop. of possible values are “at least as extreme as” observed?
 $\leadsto$ p -value!
- ▶ Compare to some α ; reject H_0 if $p < \alpha$

Randomisation Inference

- ▶ Specify and assume H_0
(sharp null of no treatment effect)
- ▶ Define H_A
- ▶ Create reference dist'n from all possible values of X under H_0
(or at least a big sample of them)
- ▶ What prop. of possible values are “at least as extreme as” observed?
 $\leadsto$ p -value!
- ▶ Compare to some α ; reject H_0 if $p < \alpha$
- ▶ CA ballot ordering effects (JASA 2006)

Randomisation Inference

The RI p -value is

$$p = \frac{\# \text{ outcomes } \geq \text{as extreme as obs}}{\text{total } \# \text{ outcomes}}$$

Randomisation Inference

The RI p -value is

$$p = \frac{\# \text{ outcomes } \geq \text{as extreme as obs}}{\text{total } \# \text{ outcomes}}$$

or

$$p = \frac{\# \text{ randomisations producing extreme } \widehat{ATE}}{\text{total } \# \text{ randomisations}}$$

Randomisation Inference

The RI p -value is

$$p = \frac{\# \text{ outcomes } \geq \text{as extreme as obs}}{\text{total } \# \text{ outcomes}}$$

or

$$p = \frac{\# \text{ randomisations producing extreme } \widehat{ATE}}{\text{total } \# \text{ randomisations}}$$

How many randomisations are there?

Combinations: Counting selected sets

How many ways to **select** k things from a set of n things?

$${}_nC_k = \binom{n}{k} = \frac{{}_nP_k}{k!} = \frac{n!}{k!(n-k)!}$$

Combinations: Counting selected sets

How many ways to **select** k things from a set of n things?

$${}_nC_k = \binom{n}{k} = \frac{{}_nP_k}{k!} = \frac{n!}{k!(n-k)!}$$

Suppose 5 units, A, B, C, D, E .

- How many ways to order? ${}_nP_k$: $ABCDE, ABCED,$
... = $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5! = 120$

Combinations: Counting selected sets

How many ways to **select** k things from a set of n things?

$${}_nC_k = \binom{n}{k} = \frac{{}_nP_k}{k!} = \frac{n!}{k!(n-k)!}$$

Suppose 5 units, A, B, C, D, E .

- ▶ How many ways to order? ${}_nP_k$: $ABCDE, ABCED, \dots = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5! = 120$
- ▶ If only want 3 of 5? Divide by $2! = (n - k)!$
(removing permutations from last 2 slots)

Combinations: Counting selected sets

How many ways to **select** k things from a set of n things?

$${}_nC_k = \binom{n}{k} = \frac{{}_nP_k}{k!} = \frac{n!}{k!(n-k)!}$$

Suppose 5 units, A, B, C, D, E .

- ▶ How many ways to order? ${}_nP_k$: $ABCDE, ABCED, \dots = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5! = 120$
- ▶ If only want 3 of 5? Divide by $2! = (n - k)!$
(removing permutations from last 2 slots)
- ▶ What if order doesn't matter? Divide by $3! = k!$
(6 permutations for ABC , but only one combination)

Combinations: Counting selected sets

How many ways to choose 5 villages of 10 for treatment?

Combinations: Counting selected sets

How many ways to choose 5 villages of 10 for treatment?

$${}_{10}C_5 = \binom{10}{5} = \frac{10!}{5!(10-5)!}$$

Combinations: Counting selected sets

How many ways to choose 5 villages of 10 for treatment?

$${}_{10}C_5 = \binom{10}{5} = \frac{10!}{5!(10-5)!}$$

$$\frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 252$$

Common Assumptions, Null Hypotheses

- ▶ Constant effect:

$$\tau_i = Y_{i1} - Y_{i0} = \tau \quad \forall i$$

- ▶ Null hypothesis of no average effect:

$$ATE = \bar{\tau} = 0$$

- ▶ Sharp null hypothesis of no effect:

$$\tau_i = 0$$

An Assignment Mechanism: Perfect Doctor

Calculate RI p -value for Perfect Doctor, under sharp null.

Patient	Y(0)	Y(1)	τ	T
1	(1)	6	(5)	1
2	(3)	12	(9)	1
3	9	(8)	(-1)	0
4	11	(10)	(-1)	0
Mean	10	9	(3)	

An Assignment Mechanism: Perfect Doctor

Calculate RI p -value for Perfect Doctor, under sharp null.

Patient	Y(0)	Y(1)	τ	T
1	(1)	6	(5)	1
2	(3)	12	(9)	1
3	9	(8)	(-1)	0
4	11	(10)	(-1)	0
Mean	10	9	(3)	

(See 02-ri-perfect-dr.R)

RI versus the t -test

Perfect Doctor:

- ▶ RI: $p = 1$
- ▶ `t.test()`: $p \approx 0.8$
- ▶ “If no tr effect, then this result typical”

RI versus the t -test

Perfect Doctor:

- ▶ RI: $p = 1$
- ▶ `t.test()`: $p \approx 0.8$
- ▶ “If no tr effect, then this result typical”

(Odd logic of NHST: “assume false thing, how strange is data?”)

Randomisation Inference, Example 2

Assumed table of potential outcomes:

Village	T	% if Female Head, $Y(1)$	% if Male Head, $Y(0)$	τ_i
1	—	15	10	5
2	—	15	15	0
3	—	30	20	10
4	—	15	20	-5
5	—	20	10	10
6	—	15	15	0
7	—	30	15	15
Average		20	15	5

Randomisation Inference, Example 2

Suppose we randomly select 2 villages to have female-headed councils, and observe

Village	T	% if Female Head, $Y(1)$	% if Male Head, $Y(0)$	τ_i
1	F	15		
2	M		15	
3	M		20	
4	M		20	
5	M		10	
6	M		15	
7	F	30		
Average		22.5	16	6.5

Randomisation Inference, Example 2

We assume the *sharp null* hypothesis (assumed values in red):

Village	T	% if Female Head, $Y(1)$	% if Male Head, $Y(0)$	τ_i
1	F	15	15	0
2	M	15	15	0
3	M	20	20	0
4	M	20	20	0
5	M	10	10	0
6	M	15	15	0
7	F	30	30	0
Average				0

Randomisation Inference, Example 2

Then we estimate what the observed ATE would be for all the possible random assignments.

First,

Village	T	% if Female Head, $Y(1)$	% if Male Head, $Y(0)$	τ_i
1	F	15		
2	M		15	
3	M		20	
4	M		20	
5	M		10	
6	F	15		
7	M		30	
Average		15	19	-4

Randomisation Inference, Example 2

Second,

Village	T	% if Female Head, $Y(1)$	% if Male Head, $Y(0)$	τ_i
1	F	15		
2	M		15	
3	M		20	
4	M		20	
5	F	10		
6	M		15	
7	M		30	
Average		12.5	20	-7.5

Randomisation Inference, Example 2

...

Randomisation Inference, Example 2

..., and all the others. The full set of $\frac{7!}{2!5!} = 21$ differences in means:

Estimate	Frequency
-7.5	3
-4	5
-0.5	6
3	2
6.5	3
10	2
Total	21

Randomisation Inference, Example 2

..., and all the others. The full set of $\frac{7!}{2!5!} = 21$ differences in means:

Estimate	Frequency
-7.5	3
-4	5
-0.5	6
3	2
6.5	3
10	2
Total	21

How many are at least as extreme as my 6.5?

Randomisation Inference, Example 2

..., and all the others. The full set of $\frac{7!}{2!5!} = 21$ differences in means:

Estimate	Frequency
-7.5	3
-4	5
-0.5	6
3	2
6.5	3
10	2
Total	21

How many are at least as extreme as my 6.5?

Two-sided (“women’s % $\neq$ men’s”): $p = \frac{8}{21} \approx 0.38$

Randomisation Inference, Example 2

..., and all the others. The full set of $\frac{7!}{2!5!} = 21$ differences in means:

Estimate	Frequency
-7.5	3
-4	5
-0.5	6
3	2
6.5	3
10	2
Total	21

How many are at least as extreme as my 6.5?

Two-sided (“women’s % $\neq$ men’s”): $p = \frac{8}{21} \approx 0.38$

One-sided (“women’s % $>$ men’s”): $p = \frac{5}{21} \approx 0.24$

Randomisation Inference

These p -values

▶ are exact, not approximate

Randomisation Inference

These p -values

- ▶ are exact, not approximate
- ▶ do not require asymptotic large $N \rightarrow \infty$

Randomisation Inference

These p -values

- ▶ are exact, not approximate
- ▶ do not require asymptotic large $N \rightarrow \infty$
- ▶ need no distrib assump (normal as $N \rightarrow \infty$)

Randomisation Inference

These p -values

- ▶ are exact, not approximate
- ▶ do not require asymptotic large $N \rightarrow \infty$
- ▶ need no distrib assump (normal as $N \rightarrow \infty$)
- ▶ do not require calculating standard error (SE)

Randomisation Inference

These p -values

- ▶ are exact, not approximate
- ▶ do not require asymptotic large $N \rightarrow \infty$
- ▶ need no distrib assump (normal as $N \rightarrow \infty$)
- ▶ do not require calculating standard error (SE)
- ▶ *Randomisation* creates distribution of possible numbers correct/treatment effects

Randomisation Inference

These p -values

- ▶ are exact, not approximate
- ▶ do not require asymptotic large $N \rightarrow \infty$
- ▶ need no distrib assump (normal as $N \rightarrow \infty$)
- ▶ do not require calculating standard error (SE)
- ▶ *Randomisation* creates distribution of possible numbers correct/treatment effects
- ▶ can be approximated by sampling

Randomisation Inference

These p -values

- ▶ are exact, not approximate
- ▶ do not require asymptotic large $N \rightarrow \infty$
- ▶ need no distrib assump (normal as $N \rightarrow \infty$)
- ▶ do not require calculating standard error (SE)
- ▶ *Randomisation* creates distribution of possible numbers correct/treatment effects
- ▶ can be approximated by sampling
 - ▶ Assigning 500 of 1000 respondents to treatment? Randomly generate 10,000 assignments from 2.7×10^{299} possible ...

Randomisation Inference

These p -values

- ▶ are exact, not approximate
- ▶ do not require asymptotic large $N \rightarrow \infty$
- ▶ need no distrib assump (normal as $N \rightarrow \infty$)
- ▶ do not require calculating standard error (SE)
- ▶ *Randomisation* creates distribution of possible numbers correct/treatment effects
- ▶ can be approximated by sampling
 - ▶ Assigning 500 of 1000 respondents to treatment? Randomly generate 10,000 assignments from 2.7×10^{299} possible ...
- ▶ can be computationally intensive

RI Example: Résumé audit study

“Causal effect of (perceived) race”?
(Bertrand and Mullainathan 2004)

```
data(resume, package = "qss")
```

	0	1
black	2278	157
white	2200	235

RI Example: Résumé audit study

“Causal effect of (perceived) race”?
(Bertrand and Mullainathan 2004)

```
data(resume, package = "qss")
```

	0	1
black	2278	157
white	2200	235

► Only possible values: $\tau_i \in \{-1, 0, 1\}$

RI Example: Résumé audit study

```
resume |> group_by(race) |>  
  summarise(call_rate = mean(call))
```

```
# A tibble: 2 x 2  
  race    call_rate  
  <chr>    <dbl>  
1 black    0.0645  
2 white    0.0965
```

- ▶ Assume the sharp null $\tau_i = 0$ for every employer.

- ▶ Assume the sharp null $\tau_i = 0$ for every employer.
- ▶ $H_0 : \mu_{\text{black name}} = \mu_{\text{white name}}$

- ▶ Assume the sharp null $\tau_i = 0$ for every employer.
- ▶ $H_0 : \mu_{\text{black name}} = \mu_{\text{white name}}$
- ▶ $H_A : \mu_{\text{black name}} \neq \mu_{\text{white name}}$

- ▶ Assume the sharp null $\tau_i = 0$ for every employer.
- ▶ $H_0 : \mu_{\text{black name}} = \mu_{\text{white name}}$
- ▶ $H_A : \mu_{\text{black name}} \neq \mu_{\text{white name}}$
- ▶ Create reference dist'n of all possible assignments

- ▶ Assume the sharp null $\tau_i = 0$ for every employer.
- ▶ $H_0 : \mu_{\text{black name}} = \mu_{\text{white name}}$
- ▶ $H_A : \mu_{\text{black name}} \neq \mu_{\text{white name}}$
- ▶ Create reference dist'n of all possible assignments

- ▶ Assume the sharp null $\tau_i = 0$ for every employer.
- ▶ $H_0 : \mu_{\text{black name}} = \mu_{\text{white name}}$
- ▶ $H_A : \mu_{\text{black name}} \neq \mu_{\text{white name}}$
- ▶ Create reference dist'n of all possible assignments

$${}_{4870}C_{2435} = \binom{4870}{2435} = \frac{4870 \cdot 4869 \cdot \dots \cdot 2436}{2435!}$$

- ▶ Assume the sharp null $\tau_i = 0$ for every employer.
- ▶ $H_0 : \mu_{\text{black name}} = \mu_{\text{white name}}$
- ▶ $H_A : \mu_{\text{black name}} \neq \mu_{\text{white name}}$
- ▶ Create reference dist'n of all possible assignments

$${}_{4870}C_{2435} = \binom{4870}{2435} = \frac{4870 \cdot 4869 \cdot \dots \cdot 2436}{2435!}$$

$$\approx 1.1 \times 10^{1464}$$

- ▶ Assume the sharp null $\tau_i = 0$ for every employer.
- ▶ $H_0 : \mu_{\text{black name}} = \mu_{\text{white name}}$
- ▶ $H_A : \mu_{\text{black name}} \neq \mu_{\text{white name}}$
- ▶ Create reference dist'n of all possible assignments

$${}_{4870}C_{2435} = \binom{4870}{2435} = \frac{4870 \cdot 4869 \cdot \dots \cdot 2436}{2435!}$$

$$\approx 1.1 \times 10^{1464}$$

(There are $\approx 10^{86}$ fundamental particles in the universe.)

- ▶ Assume the sharp null $\tau_i = 0$ for every employer.
- ▶ $H_0 : \mu_{\text{black name}} = \mu_{\text{white name}}$
- ▶ $H_A : \mu_{\text{black name}} \neq \mu_{\text{white name}}$
- ▶ Create reference dist'n of all possible assignments

$${}_{4870}C_{2435} = \binom{4870}{2435} = \frac{4870 \cdot 4869 \cdot \dots \cdot 2436}{2435!}$$

$$\approx 1.1 \times 10^{1464}$$

(There are $\approx 10^{86}$ fundamental particles in the universe.)

- ▶ Let's do 1000, or 100,000 – something reasonable

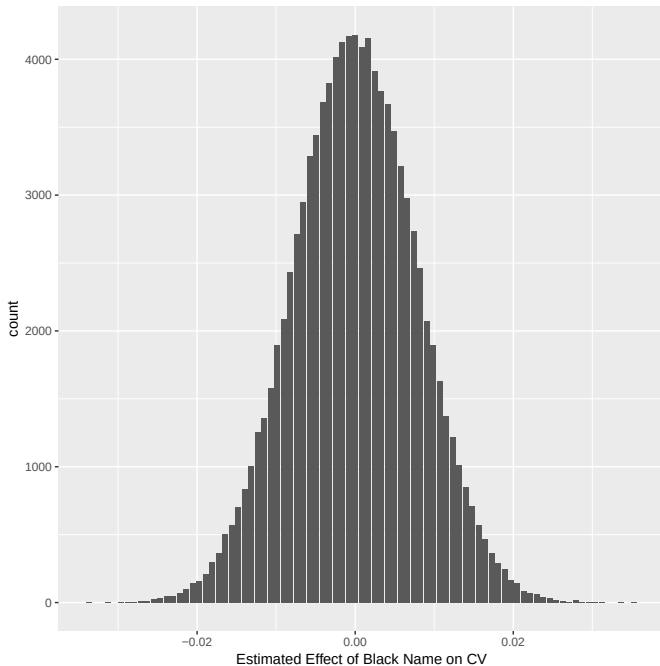
- ▶ Assume the sharp null $\tau_i = 0$ for every employer.
- ▶ $H_0 : \mu_{\text{black name}} = \mu_{\text{white name}}$
- ▶ $H_A : \mu_{\text{black name}} \neq \mu_{\text{white name}}$
- ▶ Create reference dist'n of all possible assignments

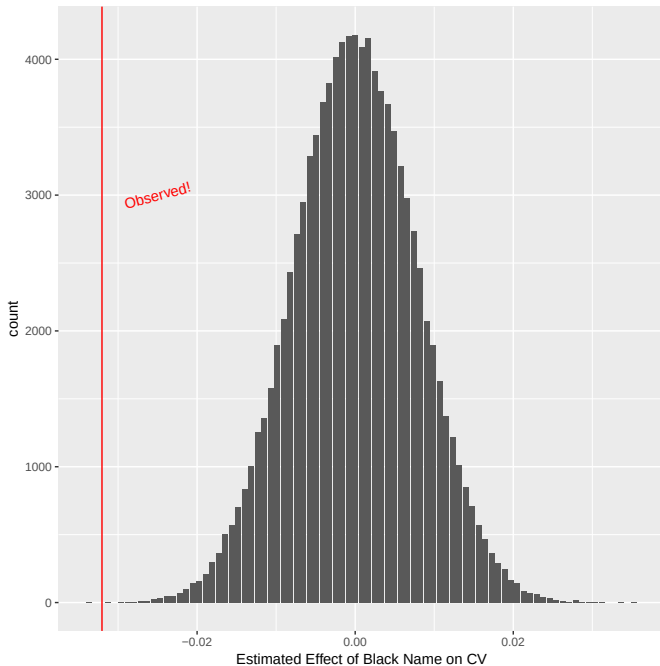
$${}_{4870}C_{2435} = \binom{4870}{2435} = \frac{4870 \cdot 4869 \cdot \dots \cdot 2436}{2435!}$$

$$\approx 1.1 \times 10^{1464}$$

(There are $\approx 10^{86}$ fundamental particles in the universe.)

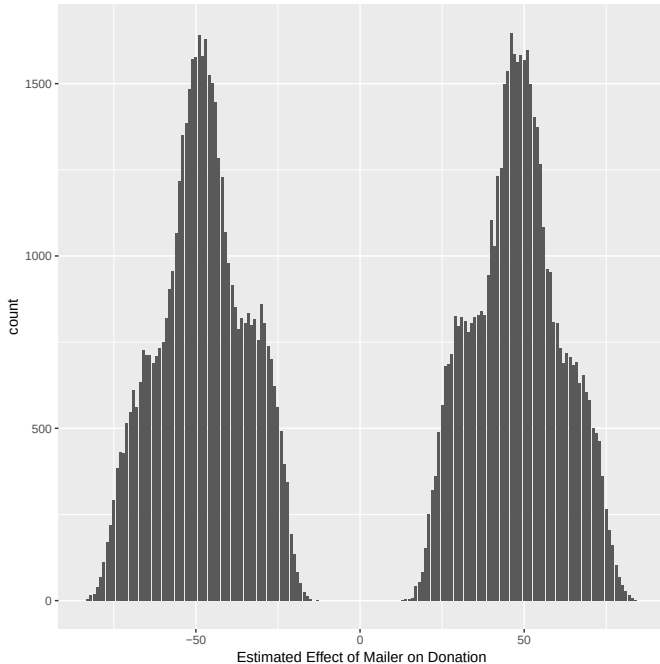
- ▶ Let's do 1000, or 100,000 – something reasonable
- ▶ See `02-ri-resume-donate.R`

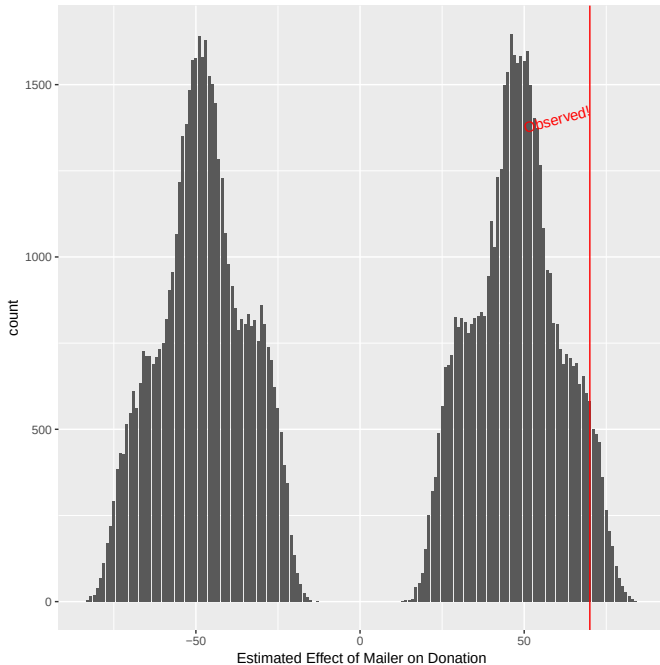




Randomisation Inference

- ▶ Gerber & Green donations example, p. 65
- ▶ Possible values $\tau_i \in (-\infty, \infty)$
- ▶ Y_1, Y_0, τ likely very skewed
- ▶ See `02-ri-resume-donate.R`





The RI Confidence Interval

Recall that

“reject H_0 at $\alpha = 0.05$ ” $\equiv$ “ H_0 falls outside 95% CI”

The RI Confidence Interval

Recall that

“reject H_0 at $\alpha = 0.05$ ” $\equiv$ “ H_0 falls outside 95% CI”

Create RI confidence ints (“invert the randomisation test”):

► Posit $H_0 : \tau = \tau^* \in \{\dots, -2, -1, 0, 1, 2, \dots\}$

The RI Confidence Interval

Recall that

“reject H_0 at $\alpha = 0.05$ ” $\equiv$ “ H_0 falls outside 95% CI”

Create RI confidence ints (“invert the randomisation test”):

- ▶ Posit $H_0 : \tau = \tau^* \in \{\dots, -2, -1, 0, 1, 2, \dots\}$
- ▶ RI test whether to reject H_0

The RI Confidence Interval

Recall that

“reject H_0 at $\alpha = 0.05$ ” $\equiv$ “ H_0 falls outside 95% CI”

Create RI confidence ints (“invert the randomisation test”):

- ▶ Posit $H_0 : \tau = \tau^* \in \{\dots, -2, -1, 0, 1, 2, \dots\}$
- ▶ RI test whether to reject H_0
- ▶ If not, then τ^* is in CI

The RI Confidence Interval

Recall that

“reject H_0 at $\alpha = 0.05$ ” $\equiv$ “ H_0 falls outside 95% CI”

Create RI confidence ints (“invert the randomisation test”):

- ▶ Posit $H_0 : \tau = \tau^* \in \{\dots, -2, -1, 0, 1, 2, \dots\}$
- ▶ RI test whether to reject H_0
- ▶ If not, then τ^* is in CI
- ▶ CI consists of set of τ^* not unusual, given data

The RI Confidence Interval

Recall that

“reject H_0 at $\alpha = 0.05$ ” $\equiv$ “ H_0 falls outside 95% CI”

Create RI confidence ints (“invert the randomisation test”):

- ▶ Posit $H_0 : \tau = \tau^* \in \{\dots, -2, -1, 0, 1, 2, \dots\}$
- ▶ RI test whether to reject H_0
- ▶ If not, then τ^* is in CI
- ▶ CI consists of set of τ^* not unusual, given data
- ▶ See `02-ri-resume-donate.R`

Additional Resources

- ▶ https://methods.egap.org/guides/analysis-procedures/randomization-inference_en.html
- ▶ https://dimewiki.worldbank.org/Randomization_Inference

Next:
Covariates in Experiments
PS2 due

References I

Bertrand, Marianne, and Sendhil Mullainathan. 2004. “Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination.” *American Economic Review* 94 (4): 991–1013.